

Manta Ray

By Michael Schratt

Illustrations by Mark McCandlish and John MacNeill

The U.S. Air Force's stealthy near real-time reconnaissance aircraft the Manta Ray evolved from a number of 1970's era classified programs aimed at developing both a deep interdiction strike fighter and a companion vehicle to gather target location data. The Air Force CSIRS (Covert Survivable In-weather Reconnaissance/Strike) program spawned two classified prototypes. One being the Lockheed Have Blue, and the other designated the Northrop THAP (Tactical High Altitude Penetrator) with McDonnell Douglas and Teledyne Ryan retained as sub-contractors. The program was jointly managed by the flight Dynamics and avionics laboratories at Wright Patterson AFB Ohio. The Air Force tested wind tunnel and RCS test models of the (THAP) span-loader concept submitted by Northrop. Northrop received a fixed price research and development/validation contract for the THAP program at the end of 1978, with the first test flight taking place in 1981 at the remote test site in Nevada. Commonly referred to as the TR-3A "Black Manta" in popular press reports, the "Manta Ray" aka "The Wraith" was used in conjunction with F-117's during Desert Storm in January of 1991 to provide digital real-time intelligence/imagery. The Manta Ray was originally based out of the Tonopah Test Range (Nevada) in 1983 along with the F-117's (although the Manta Ray was housed separately in larger hangars). Later, both squadrons were moved to Holloman AFB N.M. The Manta Ray features a flying wing configuration, with a span of 63 feet, a height of 14 feet, and a length of 42 feet. The craft is powered by two non-afterburning modified General Electric F-404 engines (rated in the 12,000 pound thrust range) which are buried deep inside the blended wing body structure, and incorporated inlets and exhaust ducts on the top of the inboard wing section. Northrop chief test pilot Paul Metz is believed to have made the first flight of this single seat aircraft.

Flight control is provided by two inward-canted vertical stabilizers (unlike the Stealth Fighter which features outward canted vertical stabilizers). The engines give the craft an un-refueled range of 3,500 miles, and special additives are mixed with the exhaust gasses to reduce the IR (Infrared Signature). With a maximum take-off weight of 45,000 pounds, the Manta Ray makes extensive use of smooth contouring curvatures (similar to the B-2), and wide use of radar absorbing materials (RAM), instead of the multiple faceted "flat plate" design incorporated in the Lockheed F-117. The Manta Ray is carrier capable (although not having any internal weapon stores), and may have taken over the role of the General Dynamics/McDonnell Douglas A-12 Avenger II (cancelled by Dick Cheney in 1991). Reports from Air Force personnel indicated that for every three F-117's, one Manta Ray built. Therefore, since a total of 59 F-117's were manufactured at Lockheed's Burbank facility under the direction of Ben Rich, it's reasonable to conclude that approximately 19 Manta Rays were constructed by late 1987.

Evidence for the existence of the Manta Ray comes from a number of different sources including: 1. A very reliable report from a KC-135 boom operator who was refueling three F-117 Nighthawks during Operation Desert Storm (January 1991). After refueling the F-117's, a single pilot bat-shaped triangular aircraft with twin inward-canted vertical stabilizers and air-intakes on top, took on fuel, and eventually broke away and joined up with the previous F-117's. 2. An eyewitness report from a fire control systems consultant, who saw a number of Manta Ray's depart the McDonnell Douglas Long Beach facility in the dead of night. The witness stated that the aircraft was just finishing up its production run, and was being deployed from the facility in the 1991-1993 time period. The witness described the classified aircraft emerging from hangars on the facility, and becoming airborne in the manner of a fully developed flight squadron. 3. A report from an eyewitness who saw the aircraft on the tarmac at the Tonopah Test Range (TTR) in 1994. 4. The classified reconnaissance aircraft has been observed flying at night with multiple F-117A's near Edwards AFB since 1989 (Aviation Week and Space Technology) June 10th, 1991).